Cloud Computing Architecture & Microservices Deployment

Course: Cloud Computing | Semester 7 | Contributor: Priya Sharma

IaaS vs PaaS vs SaaS, Docker containerization, Kubernetes pods and services, and CAP theorem.

Cloud Service Models & Virtualization

* NIST cloud characteristics: on-demand self-service, broad network access, resource pooling.
* Hypervisor Type-1 bare-metal vs Type-2 hosted virtualization.
* CAP Theorem: Consistency, Availability, Partition Tolerance trade-offs in distributed systems.

Containerization & Orchestration

* Docker namespaces and cgroups for process isolation and resource limits.
* Kubernetes architectural components: API server, etcd, scheduler, kubelet, and pods.
* Deployments, Services ClusterIP, NodePort, LoadBalancer, and Horizontal Pod Autoscaling HPA.

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